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1 Modular Arithmetic Properties

We now introduce the concept of *modular arithmetic* (also sometimes known as “clock arithmetic”). Modular arithmetic is a system of algebra in which all mathematical operations are performed relative to a *modulus* or “base”.

(Note 6, page 1) We define $x \bmod m$ (in words: “ x modulo m ”) to be the remainder r when we divide x by m . If $x \bmod m = r$, then $x = mq + r$ where $0 \leq r \leq m - 1$ and q is an integer. Explicitly,

$$x \bmod m = r = x - m \left\lfloor \frac{x}{m} \right\rfloor$$

1. Prove the following: if $a \equiv c \pmod{m}$ and $b \equiv d \pmod{m}$ then $a \cdot b \equiv c \cdot d \pmod{m}$. (Theorem 6.1 Note 6)
2. (a) If $a \equiv c \pmod{m}$ and $b \equiv d \pmod{m}$ then which of the following are true?
 - $a^b \equiv c^b \pmod{m}$
 - $a^b \equiv a^d \pmod{m}$
 - $a^b \equiv c^d \pmod{m}$
 (b) Prove your answer for part a using the theorem in question 1. If false, also provide a counterexample.
 (c) If $ka \equiv kc \pmod{m}$, does it follow that $a \equiv c \pmod{m}$?
3. Calculate $15^{2021} \pmod{17}$. (Hint: You may want to choose a different representation of 15 in mod 17.)

2 Bijections

(Note 6, Page 4) A *bijection* is a function for which every $b \in B$ has a unique *pre-image* $a \in A$ such that $f(a) = b$. Note that this consists of two conditions:

1. f is *onto*: every $b \in B$ has a pre-image $a \in A$.
2. f is *one-to-one*: for all $a, a' \in A$, if $f(a) = f(a')$ then $a = a'$.

Lemma:

For a finite set A , $f : A \rightarrow A$ is a bijection if there is an *inverse* function $g : A \rightarrow A$ such that $\forall x \in A \ g(f(x)) = x$.

1. Draw an example of each of the following situations:

One to one AND NOT onto (injective but not surjective)	Onto AND NOT one to one (surjective but not injective)	One to one AND onto (bijection, i.e. injective AND surjective)

2. Define $\mathbb{Z}_n$ to be the set of remainders mod n . In particular, $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$ for any n . Are the following functions **bijections** from $\mathbb{Z}_{12}$ to $\mathbb{Z}_{12}$?

(a) $f(x) = 7x$

(b) $f(x) = 3x$

(c) $f(x) = x - 6$

3. Why can we not have a surjection from $\mathbb{Z}_{12}$ to $\mathbb{Z}_{24}$ or an injection from $\mathbb{Z}_{12}$ to $\mathbb{Z}_6$?

4. Prove the following: The function $f(x) = a \cdot x \bmod p$ (where p is prime) is a bijection where $a, x \in \{1, 2, \dots, p-1\}$.

3 Euclid's Algorithm and Inverses

Euclid's Algorithm: Euclid's algorithm is a method to determine the greatest common factor of two numbers x and y . It hinges crucially on **Note 6, Theorem 6.3** (see question 1).

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algorithm gcd(x,y)
  if y = 0 then return(x)
  else return(gcd(y,x mod y))

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Finding Inverses with Euclid's Algorithm: Using Euclid's Algorithm, it is possible to determine the inverse of a number mod n . The inverse of x mod n is the number $x^{-1} \equiv y \pmod{n}$ such that $xy = 1 \pmod{n}$. The extended algorithm takes as input a pair of natural numbers $x \geq y$ as in Euclid's algorithm, and returns a triple of integers (d, a, b) such that $d = \gcd(x, y)$ and $d = ax + by$:

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algorithm extended-gcd(x,y)
  if y = 0 then return(x, 1, 0)
  else
    (d, a, b) := extended-gcd(y, x mod y)
    return((d, b, a - (x div y) * b))

```

1. Prove that for $a > b$, if $\gcd(a, b) = d$, then it is also true that $\gcd(b, a \bmod b) = d$. (Theorem 6.3 Note 6)
2. (a) Run Euclid's algorithm to determine the greatest common divisor of $x = 6, y = 32$.
- (b) Run Euclid's algorithm to determine the greatest common divisor of $x = 13, y = 21$. (Practice Bank, Set 4, 4c)
- (c) Use the Extended Euclid's Algorithm to find the two numbers a, b such that $13a + 21b = 1$.

(d) Given your answers to the previous parts, is there a multiplicative inverse for 13 mod 21? If so, what is it? Similarly, what is the inverse of 21 mod 13?

3. The last digit of $8k + 3$ and $5k + 9$ are the same for some k . Find the last digit of k .

4 Advanced Leapfrog

4. Suppose we have 7 vertices, each of which corresponds to a different integer modulo seven. Draw an (undirected) edge between two vertices x and y if $x + 3 \equiv y \pmod{7}$. For example, there is an edge between 0 and 3, and an edge between 5 and 2. What is the length of the shortest path between 0 and 1?

5. Suppose we have a similar setup to part 1, except now we have p vertices, for prime p , each of which corresponds to a different integer mod modulo p . Draw an edge between x and y if $x + c \equiv y \pmod{p}$. What are the possible candidates for the length of the shortest path between 0 and 1? (As this depends on the constant c and the modulus p , the answer should be in terms of modular equivalences.)

